Chapter 20 Review

Campbell Biology, AP Edition, 12e

Chapter 20 Review

A molecular model.

Summary of Key Concepts

Advanced placement edition

In bacterial transformation experiments, plasmids are typically constructed with a gene for antibiotic resistance. Explain why that is important. Explain why in some environments there is a selective advantage for bacteria with the plasmid, but in other environments, there is a selective disadvantage. **(BIG IDEAS 1 & 3)**

To review key terms, go to the **Vocabulary Self-Quiz**.

Concept 20.1: DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry

- **Nucleic acid hybridization**, the base pairing of one strand of a nucleic acid to the complementary sequence on a strand from another nucleic acid molecule, is widely used in ****DNA technology**.
- ****DNA sequencing** can be carried out using the dideoxy sequencing method in automated sequencing machines.
- Fast and inexpensive next-generation (high-throughput) techniques for sequencing DNA are based on sequencing by synthesis: DNA polymerase is used to synthesize a stretch of DNA using a single-stranded template, and the order in which nucleotides are added reveals the sequence. Third-generation sequencing methods, including nanopore technology, sequence long DNA molecules one at a time.
- ****Gene cloning** (or ****DNA cloning**) produces multiple copies of a gene (or DNA segment) that can be used to manipulate and analyze DNA and to produce useful new products or organisms with beneficial traits.
- In ****genetic engineering**, bacterial ****restriction enzymes** are used to cut DNA molecules within short, specific nucleotide sequences (****restriction sites**), yielding a set of double-

stranded ****restriction fragments with single-stranded ****sticky ends:

Diagram of DNA fragments with sticky ends.

First fragment: Top strand: 5 prime, unlabeled region, G 3 prime. Bottom strand: 3 prime, unlabeled region, C T T A A 5 prime. (Sticky end). Second fragment: Top strand: 5 prime A A T T C (sticky end), unlabeled region, 3 prime. Bottom strand: 3 prime G, unlabeled region 5 prime.

- The sticky ends on restriction fragments from one DNA source can base-pair with complementary sticky ends on fragments from other DNA molecules. Sealing the base-paired fragments with DNA ligase produces ****recombinant DNA molecules.
- DNA restriction fragments of different lengths can be separated by ****gel electrophoresis.
- The ****polymerase chain reaction (PCR) can amplify (produce many copies of) a specific target segment of DNA, using primers that bracket the desired sequence and a heat-resistant DNA polymerase.
- To clone a eukaryotic gene:

Cloning vector (often a bacterial plasmid) and DNA fragments obtained by PCR are mixed and ligated to form recombinant DNA plasmids.

Cloning vector (often a bacterial plasmid) and DNA fragments obtained by PCR or from another source (cut by same restriction enzyme used on cloning vector) are mixed and ligated to form recombinant DNA plasmids.

- Recombinant ****plasmids are returned to host cells, each of which divides to form a clone of cells.

- Expressing cloned eukaryotic genes in bacterial host cells poses several technical difficulties. The use of cultured eukaryotic cells as host cells, coupled with appropriate ****expression vectors, helps avoid these problems.

Describe how the process of gene cloning results in a cell clone containing a recombinant plasmid.

Concept 20.2: Biologists use DNA technology to study gene expression and function

- Several techniques use hybridization of a ****nucleic acid probe to detect the presence of specific mRNAs.
- ***In situ*hybridization** and ****RT-PCR can detect the presence of a given mRNA in a tissue or an RNA sample, respectively.

- Sets of genes co-expressed by a group of cells can be detected by ******RNA sequencing (RNA-seq)**—sequencing the **cDNAs** corresponding to mRNAs from the cells. ****DNA microarrays** are also used for this purpose.
- For a gene of unknown function, experimental inactivation of the gene (a gene knockout) and observation of the resulting phenotypic effects can provide clues to its function. The CRISPR-Cas9 system allows researchers to edit genes in living cells in a specific, desired way. The new alleles can be altered so that they are inherited in a biased way through a population (****gene drive**).
- In humans, ****genome-wide association studies** identify and use ****single nucleotide polymorphisms (SNPs)** as genetic markers for alleles that are associated with particular conditions.

What useful information is obtained by detecting expression of specific genes?

Concept 20.3: Cloned organisms and stem cells are useful for basic research and other applications

- The question of whether all the cells in an organism have the same genome prompted the first attempts at organismal cloning.
- Single differentiated cells from plants are often ****totipotent**: capable of generating all the tissues of a complete new plant.
- Transplantation of the nucleus from a differentiated animal cell into an enucleated egg can sometimes give rise to a new animal.
- Certain embryonic ****stem cells** (ES cells) from animal embryos and particular adult stem cells from adult tissues can reproduce and differentiate both in the lab and in the organism, offering the potential for medical use. ES cells are ****pluripotent** but difficult to acquire. Induced pluripotent stem (iPS) cells resemble ES cells in their capacity to differentiate; they can be generated by reprogramming differentiated cells. iPS cells hold promise for medical research and regenerative medicine.

Describe how, using mice, a researcher could carry out (1) organismal cloning, (2) production of ES cells, and (3) generation of iPS cells, focusing on how the cells are reprogrammed. (The procedures are basically the same in humans and mice.)

Concept 20.4: The practical applications of DNA-based biotechnology affect our lives in many ways

- DNA technology, including the analysis of genetic markers such as SNPs, is increasingly being used in the diagnosis of genetic and other diseases and in personal genome analysis. Personalized medicine offers potential for an individual minimizing their known risk for a disease, as well as better treatment of genetic disorders or cancers. ****Gene therapy** or gene editing with the

CRISPR-Cas9 system may also lead to permanent cures. DNA technology is used with cell cultures in the large-scale production of protein hormones and other proteins with therapeutic uses. Some therapeutic proteins are being produced in ****__transgenic__ "pharm" animals.
- Analysis of genetic markers such as ****__short tandem repeats (STRs)__ in DNA isolated from tissue or body fluids found at crime scenes leads to a ****__genetic profile__. Use of genetic profiles can provide definitive evidence that a suspect is innocent or strong evidence of guilt. Such analysis is also useful in parenthood disputes and in identifying the remains of crime victims.
- Genetically engineered microorganisms can be used to extract minerals from the environment or degrade various types of toxic waste materials.
- The aims of developing transgenic plants and animals are to improve agricultural productivity and food quality.
- The potential benefits of genetic engineering must be carefully weighed against the potential for harm to humans or the environment.

What factors affect whether a given genetic disease would be a good candidate for successful gene therapy?

*For suggested answers, see*Appendix A.**

Test Your Understanding

For more multiple-choice questions, go to the **Practice Test**.

Levels 1-2: Remembering/Understanding

- In DNA technology, the term *vector* can refer to
- the enzyme that cuts DNA into restriction fragments.
- the sticky end of a DNA fragment.
- a SNP marker.
- a plasmid used to transfer DNA into a living cell.
- Which of the following tools of DNA technology is incorrectly paired with its use?
- electrophoresis—separation of DNA fragments
- DNA ligase—cutting DNA, creating sticky ends of restriction fragments
- DNA polymerase—polymerase chain reaction to amplify sections of DNA

- reverse transcriptase—production of cDNA from mRNA
- Plants are more readily manipulated by genetic engineering than are animals because
- plant genes do not contain introns.
- more vectors are available for transferring recombinant DNA into plant cells.
- a somatic plant cell can often give rise to a complete plant.
- plant cells have larger nuclei.
- A paleontologist has recovered a bit of tissue from the 400-year-old preserved skin of an extinct dodo (a bird). To compare a specific region of the DNA from a sample with DNA from living birds, which of the following would be most useful for increasing the amount of dodo DNA available for testing?
- SNP analysis
- polymerase chain reaction (PCR)
- electroporation
- gel electrophoresis

Levels 3-4: Applying/Analyzing

- Which of the following is true of cDNA produced using human brain tissue as the starting material?
- The procedure to make it requires amplification by the polymerase chain reaction.
- It is produced from pre-mRNA using reverse transcriptase.
- It can be labeled and used as a probe to detect genes expressed in the brain.
- It includes the introns of the pre-mRNA.
- Expression of a cloned eukaryotic gene in a bacterial cell involves many challenges. The use of mRNA and reverse transcriptase is part of a strategy to solve the problem of
- post-transcriptional processing.
- post-translational processing.
- nucleic acid hybridization.

- restriction fragment ligation.
- Which of the following sequences in double-stranded DNA is most likely to be recognized as a cutting site for a restriction enzyme?
- **A.** AAGGTTCC
- **B.** GGCCCCGG
- **C.** ACCATGGT
- **D.** AAAATTTT

Levels 5-6: Evaluating/Creating

AP Assessment Questions

- **CONNECT TO BIG IDEA 3** Imagine you want to study one of the human crystallins, proteins present in the lens of the eye (see Figure 1.8). To obtain a sufficient amount of the protein of interest, you decide to clone the gene that codes for it. Assume you know the sequence of this gene. **Explain** how you would go about this.
- **CONNECT TO BIG IDEA 3** Looking at Figure 20.15, what does it mean for a SNP to be "linked" to a disease-associated allele? **Explain** how this allows the SNP to be used as a genetic marker. (See Concept 15.3.)
- **DRAW IT *Science Practice 2*** You are cloning an aardvark gene, using a bacterial plasmid as a vector. The green diagram shows the plasmid, which contains the restriction site for the enzyme used in Figure 20.5. Above the plasmid is a segment of linear aardvark DNA that was synthesized using PCR. **Diagram** your cloning procedure, and **show** what would happen to these two molecules during each step. Use one color for the aardvark DNA and its bases and another color for those of the plasmid. **Label** each step and all 5′ and 3′ ends.

Aardvark DNA segment and plasmid which contains the restriction site for the enzyme used in Figure 20.5.

Two tables are given. First:

5 prime	**G**	**A**	**A**	**T**	**T**	**C**	**T**	**A**	**A**	**A**	**G**	**C**	**G**	**C**	**T**
3 prime	C	T	T	A	A	G	A	T	T	T	C	G	C	G	A

Second:

Top Strand (outer circle)GAATTCBottom Strand (inner circle)CTTAAG

- **CONNECT TO BIG IDEA 1** Ethical considerations aside, if DNA-based technologies became widely used, **discuss** how they might change the way evolution proceeds, as compared with the natural evolutionary mechanisms that have operated for the past 4 billion years.
- **SCIENTIFIC INQUIRY** ***Science Practice 3***You hope to study a gene that codes for a neurotransmitter protein produced in human brain cells. You know the amino acid sequence of the protein. **Explain** how you might (a) identify what genes are expressed in a specific type of brain cell, (b) identify (and isolate) the neurotransmitter gene, (c) produce multiple copies of the gene for study, and (d) produce large quantities of the neurotransmitter for evaluation as a potential medication.
- **CONNECT TO BIG IDEA 3** In a short essay (100–150 words), **discuss** how the genetic basis of life plays a central role in biotechnology.
- **SYNTHESIZE YOUR KNOWLEDGE*****Science Practice 6***

Steam rises from a hot spring in Yellowstone National Park.

- The water in the Yellowstone National Park hot springs shown here is around 160°F (70°C). Biologists assumed that no species of organisms could live in water above about 130°F (55°C), so they were surprised to find several species of bacteria there, now called *thermophiles* ("heat-lovers"). You've learned in this chapter how an enzyme from one species, *Thermus aquaticus*, made feasible one of the most important DNA-based techniques used in labs today. **Identify** the enzyme, and **indicate** the value of its being isolated from a thermophile. **Suggest** other reasons why enzymes from this bacterium (or other thermophiles) might also be valuable.

*For selected answers, see*Appendix A.**